# SEC 2-FA2 GROUP4-EDUVALA, MIKHAEL

GITHUB - Probability and Probability Distribution

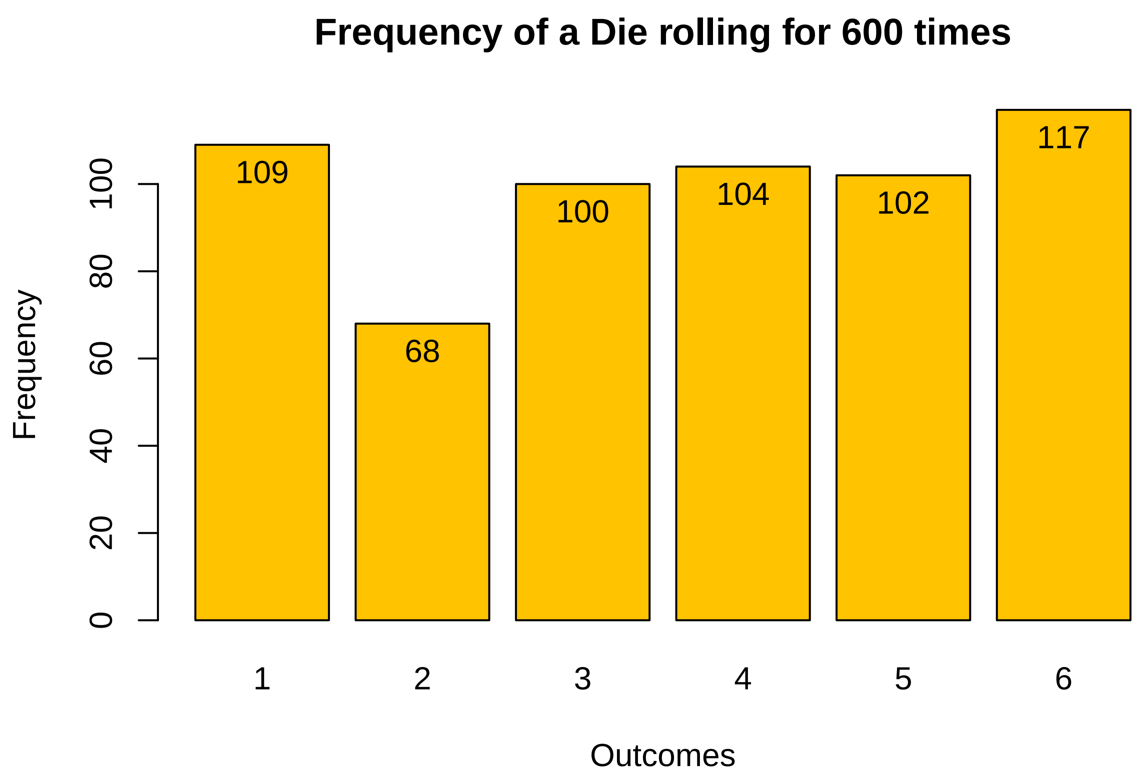
2026-08-30

### 3. An experiment consists of rolling a die. Use R to simulate this experiment 600 times and obtain the relative frequency of each possible outcome. Hence, estimate the probability of getting each of 1, 2, 3, 4, 5, and 6.

```
x <- c(1,2,3,4,5,6)

experiment <- sample(x, 600, replace = TRUE)

frequency <- table(experiment)
print(frequency)
```

```
## experiment
##   1   2   3   4   5   6
## 109  68 100 104 102 117
```

```
bar <- barplot(frequency, names.arg = x, main = "Frequency of a Die rolling for 600 times",
               xlab = "Outcomes", ylab = "Frequency", ylim = c(0,max(frequency)),col="#FFC300")

text(bar, frequency, labels = frequency, cex = 1, pos = 1)
```




##Shown above is the frequency of a die getting 1,2,3,4,5,6 for 600 times

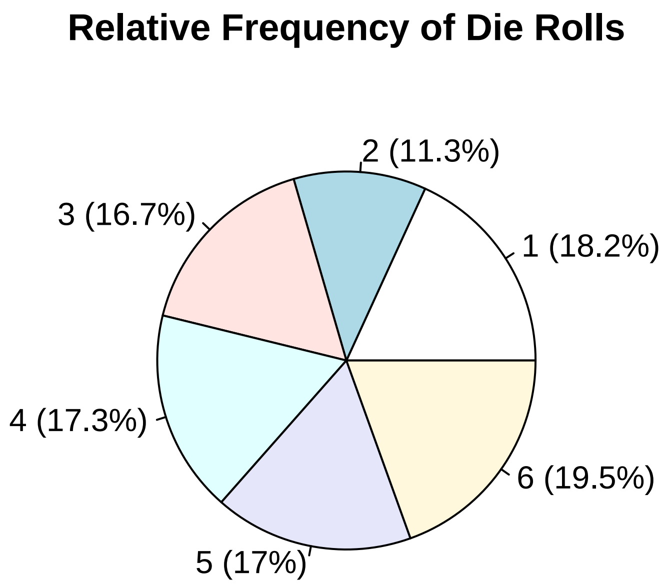
### Relative Frequency:

Here is the estimated probability of getting each 1,2,3,4,5,6 in a die.

```
relative_frequency <- frequency / 600
print(relative_frequency)
```

```
## experiment
##         1         2         3         4         5         6
## 0.1816667 0.1133333 0.1666667 0.1733333 0.1700000 0.1950000
```

```
pie(
  relative_frequency,
  labels = paste0(
    names(relative_frequency),
    " (", round(relative_frequency * 100, 1), "%)"
  ),
  main = "Relative Frequency of Die Rolls"
)
```




##The Pie Chart above shows the relative frequency of getting 1,2,3,4,5,6 in a die rolling for 600 times